INTERIM FEDERAL SPECIFICATION

SIGN, INTERIOR ILLUMINATED, ELECTRIC, PORTABLE

(WITH INTERCHANGEABLE SIGN FACE)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington 25, D. C., based upon currently available technical information. It has been partially coordinated with industry, but has not been coordinated with government agencies and is subject to modification. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for an interior illuminated portable flashing sign with interchangeable sign face (including legend).

1.2 Classification.

1.2.1 Types, classes and styles. Interior illuminated signs shall be of the following types, classes and styles, as specified (see 6.2).

Type I - Sign, Interior Illuminated, Electric, with Photocell.
Type II - Sign, Interior Illuminated, Electric, without Photocell.

Class 1 - With brackets (for mounting on conventional "horses").
Class 2 - With pipe mount (for mounting on stanchions).
Class 3 - Without brackets or pipe mount for mounting.

Style 1 - With carrying handle.
Style 2 - Without carrying handle.

1.2.2 Sizes and legends. Sizes and legends shall be as specified in the contract or order (see 3.6 and 6.2).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Box, Fiberboard.

FSC 9905

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 406 - Plastics: Methods of Testing.

FOR INSTRUCTIONS ON OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS, SEE LAST PAGE.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Manual on Uniform Traffic Control Devices for Streets and Highways.

Code of Federal Regulations, Title 10, Section 20.203.

(Application for copies may be obtained by addressing the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

American Society for Testing and Materials Standards:

D256-56 - Impact Resistance of Plastics and Electrical Insulating Materials.
D635-56T- Flammability of Rigid Plastics Over 0.050 In. in Thickness.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Standard product. Interior illuminated signs covered by this specification shall be standard products of the manufacturer except for such deviations from the manufacturer's standards as may be required by this specification.

3.2 Figures. Figures incorporated in this specification are included for illustration purposes only and are not intended to preclude signs otherwise meeting the requirements of this specification.

3.3 Preproduction sample. When specified (see 6.2), before production is commenced, a sample of the finished item covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval. The approval of the preproduction sample does not

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.

relieve the contractor of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production items.

3.4 Standards for traffic control and radiation caution signs.

3.4.1 Traffic control signs. Traffic control signs shall conform to all requirements of the Manual on Uniform Traffic Control Devices for Streets and Highways.

3.4.2 Radiation caution signs. Radiation caution signs shall conform to the standards prescribed in the Code of Federal Regulations, Title 10, Section 20.203.

3.5 Design and construction. The interior illuminated sign (see figures 1 and 2) shall consist of an opaque vacuum formed or molded plastic (see 3.7.1.1) sign case and securely attached translucent plastic sign face with legend. The sign shall have a manual switch control, with photocell, type I, or without photocell, type II, as specified (see 6.2) and shall be designed to contain batteries. The sign shall be protected where the sign case and sign face are joined together to prevent moisture or dirt from reaching the interior of the sign. This can be accomplished by use of a rubber or plastic gasket or similar method. The sign shall be of such weight and construction as to withstand reasonable shock without breakage or other damage to exterior and interior components and shall be constructed to stand firmly without supports in an upright position on a level surface with batteries inserted. Class 3, style 1 signs, size 12 x 24 inches, shall weigh completely assembled not less than 11 lbs. with batteries inserted. This weight shall be used as a guide to ensure that signs in other sizes and with varying accessories conform to reasonable minimum weight limits.

3.5.1 Exterior of sign. There shall be a removable panel at the back of the sign case to enable change in batteries and circuit adjustment. The sign shall be furnished with brackets, class 1, on back of sign for mounting on conventional "horses"; pipe mount, class 2, on back of sign for mounting on stanchions; or without mounting brackets or pipe mounts, class 3. The sign shall be furnished with a carrying handle, style 1, or without a carrying handle, style 2, as specified (see 6.2).

3.5.2 Interior of sign. The interior of the sign shall consist of tube, power booster, variable resistor, circuit wiring and parts necessary to complete electrical assembly, and brackets and supports to mount or secure interior components. To prevent breakage or damage to the tube, insulated cushioning material shall be used with or on brackets or supports, and other areas where the tube would be in contact with hard surfaces, for protection against reasonable shock or impact.

3.5.3 Sign face. The sign face specified in 3.5 shall be made of plastic as specified in 3.7.1.2.

The background color and legend shall be as specified in the contract or order (see 6.2). Unless otherwise specified by the contracting officer (see 6.2), traffic control (including construction warning) signs shall have vacuum formed letters or symbols and radiation caution signs shall have flat face letters or symbols.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.

Size and color of letters or symbols shall be as furnished by the contractor and shall conform to requirements specified in 3.4. When sign faces are procured or ordered separately, the size, shape if other than rectangular, legend and type of attachment (e.g. number and location of holes, when applicable) shall be as specified by the contracting officer (see 6.2). Sign faces of the same size and shape shall be interchangeable with all sign cases of the corresponding size and shape.

3.6 Size and shape. Size of the sign shall be as specified in the contract or order (see 6.2). The size shall be designated by dimensions of the sign face (e.g. length and width for rectangular signs). Unless otherwise specified (see 6.2), the shape of the sign and illuminated area shall be rectangular.

3.7 Materials. Materials shall be as hereinafter specified. Materials not specifically defined shall be as regularly used by the manufacturer.

3.7.1 Plastic.

3.7.1.1 Sign case. The plastic used for the sign case shall have high impact strength and abrasion resistance, with low moisture absorption property, not to exceed slow burning classification (ASTM flammability test method D635-56T), and shall meet the following requirements:

Flexural strength (psi)(min.) 1/		- 6000
Impact strength (ft. lbs./inch notch) 2/	70° F.	- 9.0
	-20° F.	- 2.0
Tensile strength (psi)(min.) 3/		- 3800

1/ Test method 1031 of Fed. Std. No. 406.
2/ ASTM method D256-56.
3/ Test method 1011 of Fed. Std. No. 406.

3.7.1.2 Sign face. The plastic used for the sign face shall have the following properties: low moisture absorption, high distortion temperature, be resilient, and slow burning (ASTM flammability test method D635-56T). The thickness shall be not less than .060 inches.

3.7.2 Photocell (type I). When type I sign with photocell is specified, the photocell shall conform to the following.

(a) in operation draw between 5 and 10 milliamperes of current from batteries.
(b) in off position draw almost no measurable amount of current.
(c) shall have wire leads and be sealed against moisture and dust.
(d) overall dimensions approximately 1" x 2", excluding mounting legs.
(e) be factory adjusted so that sign will be operative from dusk until dawn, unless otherwise specified (see 6.2).

3.7.3 Tube. Neon, argon, mercury vapor, krypton or combination thereof, with tube phosphors to enhance readability and application. The tubes shall be designed and made so that they will function for the life of the sign (except for conditions such as breakage or damage).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.

3.7.4 Power booster. Power booster shall be encapsulated for environmental protection. The booster shall be matched to the tube (size, gas pressure and content) for maximum efficient performance.

3.7.5 Hardware. Hardware shall be made of corrosion resistant metal, or painted or otherwise treated to resist corrosion.

3.8 Operation. The sign shall operate with two 6-volt alkaline cell batteries, unless otherwise required, providing a minimum 750 hour continuous flashing operation. When equipped with photocell and adjusted to operate from dusk until dawn, the minimum life of the batteries shall be 1500 hours. Unless otherwise specified (see 6.2), batteries shall not be furnished with the sign. When the sign is in operation, illumination shall be such that flashing of the sign at night under clearly visible conditions can be seen at a distance of at least 1/4 mile. Unless otherwise specified by the contracting officer (see 6.2), the flash rate shall be adjusted to 40 (± 10) flashes per minute. Unless otherwise specified by the contracting officer, the flash duration shall be a minimum of 100 milliseconds.

3.8.1 Instructions. Instructions shall be included on the removable panel by decal, or otherwise, indicating the type of batteries to be used and assembly instructions.

3.9 Workmanship. The sign shall be free from defects such as cracks, damaged components, peeling paint or any other defects which would affect serviceability of the unit.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Preproduction sample inspection. When required, inspection of the preproduction sample shall be made of a completed interior illuminated sign for all the provisions of this specification applicable to end product examination and tests.

4.2 Inspection for acceptance. Inspection shall be performed in accordance with the provisions set forth in Standard MIL-STD-105, except where otherwise indicated.

4.2.1 Certificate of compliance. The contractor shall submit a certificate of compliance to the contracting officer certifying that all components used in the manufacture of the interior illuminated sign conform to the requirements of this specification. The contractor shall further certify that the signs furnished are standard products of manufacture, except as otherwise required by this specification, and based on reliability tests performed (sampling or otherwise) conform to all requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.

4.2.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process or assembly operations to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material on which in-process inspection has indicated nonconformance.

4.2.3 Examination of end item. The sign shall be examined for the following defects in design, material, construction, and workmanship. The lot for purposes of examination shall be the signs offered for delivery. For sampling, the inspection level shall be level I of Standard MIL-STD-105 and acceptable quality level (AQL) expressed in defects per hundred units shall be 1.0. The unit of product for this examination shall be one completely assembled interior illuminated sign. The Government reserves the right to perform 100 percent inspection of submitted lots.

Defect
Material not as specified.
Electrical and circuit assembly not complete.
Mountings not as specified.
Any specified component missing.
Interior components not adequately mounted or secured with brackets or supports.
Insulated cushioning material not used to protect the tube from contact with hard surfaces.
Interior of sign not adequately protected against entry of moisture or dirt.
Sign cannot stand firmly in upright position on level surface.
Legend not as specified.
Background color not in accordance with that specified.
Sign face not secure and evenly attached.
Painted or silkscreened areas peeling or with bare spots.
Instructions not included on removable panel.
Size not as specified.

4.3 Operational test. Each sign, with batteries installed, shall be operated by turning the switch on and off. With the switch in the "ON" position, the lamp should light and illuminate the sign. Failure of the switch to turn the lamp on and off or failure of the illuminated area of the sign to be seen by the unaided eye at night under clearly visible conditions at a distance of at least 1/4 mile shall constitute failure of this test. Batteries for the test shall be furnished by the contractor when the test is performed at or near the contractor's plant facilities.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian procurements.)

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.

5.1.1 Level A or B.

5.1.1.1 Units. Each complete unit shall be individually packaged in a box conforming to PPP-B-636, type I or II, minimum bursting strength 200 p.s.i. The unit shall be cushioned with cellulosic pads, macerated paper pads or built-up fiberboard pads. The material shall be secured in place with tape. Fiberboard blocking and end pads taped over each corner of the face, shall effect a floating suspended type pack.

5.1.1.2 Individual sign faces. Individual faces shall be wrapped in kraft paper. If more than one, the faces shall be stacked with separators of glassine or kraft paper. Faces shall be packaged in a box conforming to PPP-B-636, type I or II, minimum bursting strength 200 p.s.i.

5.1.2 Level C. Signs and faces shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Signs or faces, packaged as specified in 5.1, shall be packed in a box conforming to PPP-B-636, type I or II, class 2, minimum bursting strength 200 p.s.i. The box shall be securely sealed, with adhesive applied throughout the entire area of contact between the flaps, and waterproof sealed with tape in accordance with the appendix to the box specification.

5.2.2 Level B. Signs or faces, shall be packed as specified in 5.2.1, except that the box shall be type I, class 1, and waterproofing shall not be required. Sealing and closure shall be in accordance with the box specification.

5.2.3 Level C. Signs or faces shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the interior and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The interior illuminated sign is intended for use primarily as a portable unit for traffic control, road marking, warning and radiation caution signs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and style (see 1.2.1).
(c) Size of sign (see 3.6 and 6.2.1).
(d) Legend (see 3.5.3 and 6.2.1).
(e) Background color of sign face (see 3.5.3 and 6.2.2).
(f) When preproduction sample is required (see 3.3).
(g) Applicable levels of packaging and packing (see 5.1 and 5.2).
(h) Shape, if other than specified (see 3.6).
(i) Adjustment for photocell, if other than specified (see 3.7.2 and 6.2.3).
(j) Flash rate, if other than specified (see 3.8).
(k) Form (vacuum formed or flat face) of letters or symbols, if other than specified (see 3.5.3 and 6.2.4).
(l) When sign faces are procured or ordered separately, the size, legend, shape if other than rectangular, and type of attachment (see 3.5.3).

6.2.1 Legend and size of signs. For the guidance of procurement officers, table I lists some of the legends and sizes of signs which are available for procurement. Signs procured or ordered should not necessarily be restricted to the legends or sizes listed, which are given for information only.

6.2.2 Background color for sign face. Traffic control warning signs generally have a yellow background and traffic control regulatory signs generally have a white background, in accordance with the Manual on Uniform Traffic Control Devices for Streets and Highways. Radiation caution signs generally have a yellow background, in accordance with the Code of Federal Regulations, Title 10, Section 20.203.

6.2.3 Photocell. Photocells on type I signs are factory adjusted by the contractor to be operative from dusk until dawn, as specified in subparagraph (d) of 3.7.2. However, photocells can be made to operate continuously without automatic shutoff at dawn, if required.

6.2.4 Vacuum formed or flat face letters and symbols. The form (vacuum formed or flat face) used for letters or symbols is based on the length of the message to facilitate reading of the legend. Traffic control signs generally have the shorter messages and therefore vacuum forming is used, whereas radiation caution signs generally have the longer messages and flat face is used. The length of the message ultimately governs the form to be used.

6.3 Batteries. Batteries should not normally be procured or ordered with the sign (see 3.8), because of increased transportation costs due to added weight of the batteries and possible damage in transit. Batteries required are generally available locally.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.

TABLE I

Traffic Control and Construction Signs:

Symbol or Legend	Size
STOP AHEAD	24 x 24"
ONE WAY (Specify right or left arrow)	12 x 36"
ONE WAY (Specify right or left arrow)	18 x 24"
SPEED LIMIT (Specify speed)	18 x 24"
END (MPH) MILE SPEED	18 x 24"
STOP (red)	24 x 24"
STOP (red)	30 x 30"
YIELD	30" triangle
YIELD	36" triangle
NO RIGHT TURN	18 x 24"
NO LEFT TURN	18 x 24"
NO U TURN	18 x 24"
DO NOT PASS	18 x 24"
PASS WITH CARE	18 x 24"
KEEP RIGHT (With right arrow)	18 x 24"
DO NOT ENTER	24 x 24"
NO PARKING ANYTIME	12 x 18"
NO PARKING BETWEEN SIGNS	12 x 18"
NO PARKING LOADING ZONE	12 x 18"
ACCIDENT AHEAD	12 x 24"
NO PARKING BUS STOP	12 x 18"
PARKING DIAGONAL	12 x 18"
RADAR ENFORCED	12 x 18"
DIP	24 x 24"
ROAD NARROWS	30 x 30"

Large Construction Signs:

Symbol or Legend	Size
MEN WORKING AHEAD	30 x 30"
ROAD CLOSED	30 x 30"
DETOUR (Specify right or left arrow)	30 x 30"
STOP AHEAD	30 x 30"
MERGING TRAFFIC	30 x 30"
ROAD REPAIRS AHEAD	30 x 30"
MEN WORKING	30 x 30"

Radiation Caution Signs:

Symbol or Legend	Size
DANGER CONTAMINATED WATER	24 x 12"
RADIATION DANGER HAZARD	12 x 24"
CLOSED AREA AUTHORIZED PERSONNEL ONLY	36 x 12"
RADIATION AREA AUTHORIZED PERSONNEL ONLY	36 x 12"

Notice: (See last page).

GSA-WASH DC 64-1764

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.



FIGURE 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.

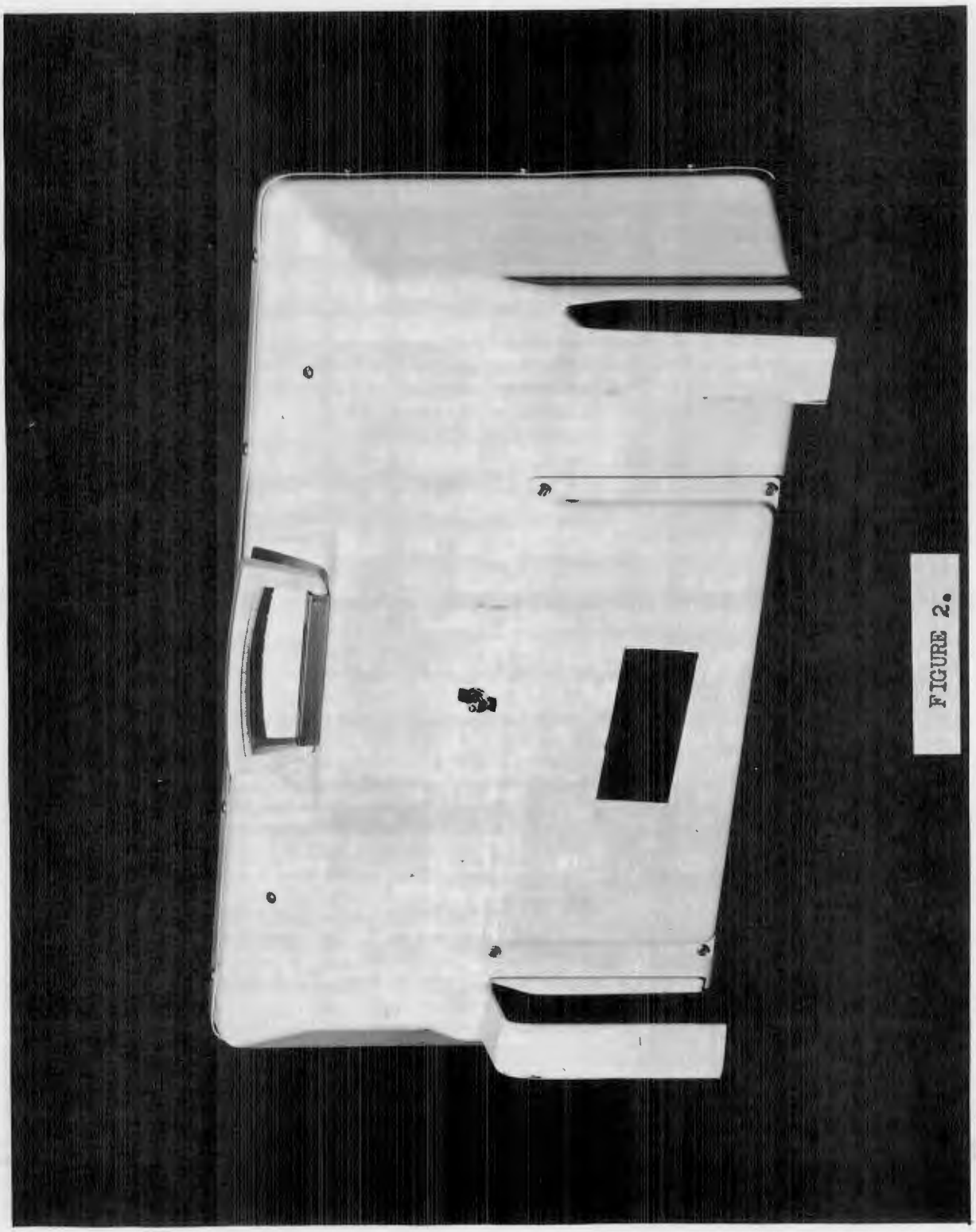


FIGURE 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS ON OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards and Handbooks from established distribution points in their agencies.)

NOTICE

When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

GSA WASH DC 62-13090

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:09Z
Check the source to verify that this is the current version before use.